

Inchcape Demand Project

Collision Parts Forecasting

UNIVERSITY OF CAMBRIDGE
DATA SCIENCE WITH MACHINE LEARNING & AI CAREER
ACCELERATOR



D8 Solutions
DATA SCIENCE

JOVAN SURYA

ANONYMISED PEER 1

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ANONYMISED PEER 3

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ANONYMISED PEER 5

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Background and Business Context

Business problem and project objective

Inchcape's aftersales spare-parts operation supports vehicle availability, repair-centre efficiency and customer retention. For Suzuki Chile, the business problem was to forecast monthly collision-part demand at SKU level. Availability is especially important for collision repairs: a missing component can keep a vehicle off the road, delay insurer-authorised work and increase the risk that future repairs are redirected elsewhere.

Collision demand is more difficult to predict than routine maintenance demand because it is sparse, volatile and event driven. Many SKUs have long zero-demand periods followed by isolated spikes. Under-forecasting therefore creates stockouts and delays, whereas over-forecasting locks working capital into slow-moving or obsolete stock. The existing collision-parts benchmark of approximately 70% WMAPE also indicated substantial scope for improvement.

The project aimed to turn historical SKU-level sales into a repeatable 18-month forecasting workflow. The meaningful solution was not a single universal model, but a decision-support system that segments demand, compares appropriate models fairly, selects the strongest method for each segment and demand-volume tier, and produces traceable forecasts for planners.

Project Development Process

Data preparation and problem segmentation

The source contained 952,504 monthly SKU records. Checks on the SKU identifier, month, sales volume and collision flag found no duplicates or missing values. Filtering produced 8,097 collision SKUs (42% of all SKUs). A complete monthly panel was then created, explicitly filling unobserved months with zero rather than treating them as missing; this distinction was essential for modelling intermittent demand.

History length varied from 1 to 64 months (median 58). Because short histories weaken training and backtesting, eligibility was based on having sufficient observations before each forecast origin. This avoided presenting unreliable comparisons for recently introduced or rarely observed SKUs; coverage diagnostics are reported in Appendix A.

Demand was highly concentrated: the top 5% of SKUs generated 80% of volume and the top 20% generated 95%. Consequently, aggregate accuracy alone could hide poor performance on commercially important parts. The evaluation was therefore stratified across all eligible SKUs, the top 20% and the top 5% by demand.

SKUs were classified using average demand interval (ADI) and squared coefficient of variation (CV^2). Intermittent items represented 81.4% of SKUs but only 22.2% of demand, while smooth items represented 12.0% of SKUs and 65.8% of volume (Table 1). This finding shaped the solution: sparse demand required specialist methods, but high-volume smooth items deserved separate commercial weighting.

Type	SKU's	Demand (total)	% zero months	% SKU	% demand
Intermittent	6593	52,809	83.3%	81.4%	22.2%
Smooth	972	156,317	7.1%	12.0%	65.8%
Lumpy	381	12,349	63.6%	4.7%	5.2%
Erratic	124	16,049	11.6%	1.5%	6.8%
No demand observed	27	0		0.3%	0%

Table 1. Demand-type distribution among collision SKUs

Model portfolio and rationale

A manual experimentation phase first compared complementary model families rather than committing to one technique. ARIMA and related statistical methods provided transparent time-series benchmarks; Croston SBA addressed intermittent demand explicitly; XGBoost and LightGBM captured nonlinear relationships from lag, rolling and calendar features; LSTM tested shared sequential patterns; and hybrid ensembles tested whether temporal and feature-based signals were complementary. Chronos and Moirai were included as zero-shot foundation models because they can transfer learned time-series representations without SKU-specific training.

Every candidate was compared with a three-month moving-average baseline. WMAPE was the primary measure because it weights errors by actual demand and therefore reflects operational exposure better than an unweighted average of SKU errors. Model performance was also examined by demand type, demand-volume tier, forecast horizon, stability across origins, and the share of SKUs below the 70% benchmark. These complementary views prevented a model from being selected on one favourable aggregate statistic.

The manual notebooks exposed two refinements needed for reliability. First, different notebook structures and output schemas made direct comparison fragile, so preprocessing, metrics and result files were standardised. Second, selected SKU forecasts were visually inspected for implausible shapes, negative values and poor handling of zero-demand periods. The manual phase therefore became a proof of concept and a requirements definition for the automated pipeline rather than the final production process.

The resulting design principle was consistent evaluation before optimisation: all models would receive the same zero-filled panel, forecast origins and lag assumptions, and would return a common set of predictions and diagnostics.

Automated forecasting solution

Pipeline design and reproducibility

The proof of concept became a GitHub-based, 14-agent orchestrator. A full run follows six controlled stages:

1. Validate the raw data and construct the zero-filled monthly demand panel.
2. Classify SKUs using ADI and CV², then engineer lag, rolling, sparsity and seasonal features.
3. Run all registered models on identical rolling-origin back tests.
4. Compare WMAPE by demand type and volume tier and select the validated champion.
5. Generate the 18-month forecast with the selected model.
6. Retain diagnostics, configuration and outputs for audit and reproducibility.

The complete sequence and model registry are shown in Appendices G–H.

Validation safeguards

Fairness was enforced by using identical forecast origins, a 12-month back test horizon, a three-month information lag and the same aggregation logic for every model. No future observations were used as features. For ensembles, blending weights were tuned on an inner training slice rather than the evaluation period, reducing leakage. Repeated origins tested temporal stability, while horizon profiles showed whether performance deteriorated as forecasts extended. Cached outputs and configuration files made every comparison reproducible.

Iteration, retraining and adaptability

The pipeline supports rapid iteration through two configurations: a capped demonstration mode for debugging and a full mode for production evaluation. Agents can be rerun individually using validated cached outputs, allowing a new model or metric to be assessed without repeating every expensive stage. Dataset paths, column names, model parameters and selection metrics are exposed in YAML, so the same workflow can be retrained on new monthly data or adapted to another market or vehicle brand with the same SKU–date–demand schema.

Results

Backtest findings

The common backtest confirmed that forecastability depends on both demand behaviour and volume. Smooth demand produced the lowest WMAPE, whereas intermittent and lumpy demand remained above the 70% benchmark even for the leading methods. Across all backtested SKUs, Chronos2 Small achieved the lowest overall WMAPE (approximately 61%) and won the intermittent and lumpy segments; Moirai performed best for smooth and erratic demand. The difference from the earlier notebook-level comparison demonstrates why final recommendations must be based on the standardised pipeline rather than separately generated outputs.

Commercial-volume findings

Accuracy improved as evaluation focused on commercially important items. Overall WMAPE decreased from approximately 61% across all eligible SKUs to 56% for the top 20% and 48% for the top 5%. The winning method nevertheless changed by tier and segment, confirming that demand volume must form part of the selection rule. Table 2 summarises the overall winners.

Evaluation tier	Overall winner	WMAPE	Decision implication
All eligible SKUs	Chronos2 Small	≈61%	Best broad portfolio benchmark
Top 20% by demand	Chronos2 Small	≈56%	Priority high-volume portfolio
Top 5% by demand	Croston	≈48%	Most commercially critical SKUs

Table 2. Overall winner by demand-volume tier

Limitations and monitoring

The results should be interpreted within the available 64-month history. Short-history SKUs were less likely to qualify for backtesting, and collision demand can be affected by shocks not represented in sales alone. Moreover, WMAPE measures forecast error but does not directly optimise stockout cost, holding cost or service level. In deployment, realised demand and error should therefore be monitored monthly by segment, with champion models reconsidered when accuracy drifts. Future iterations should test validated external drivers and incorporate lead times and inventory costs into the final ordering rule.

Recommended solution and business use

The recommended solution is a configurable champion-by-segment approach. Operational use should follow five steps:

1. Rerun the common backtest during each scheduled retraining cycle.
2. Select the best validated model within each demand-type and volume tier instead of hard-coding today's winner.

3. Prioritise the top 5% and top 20% of SKUs for planner review because they drive most inventory exposure.
4. Apply uncertainty and stock-policy safeguards to intermittent, lumpy and short-history items.
5. Combine forecasts with lead times, service-level targets and business knowledge, then trigger reselection when monitored accuracy deteriorates.

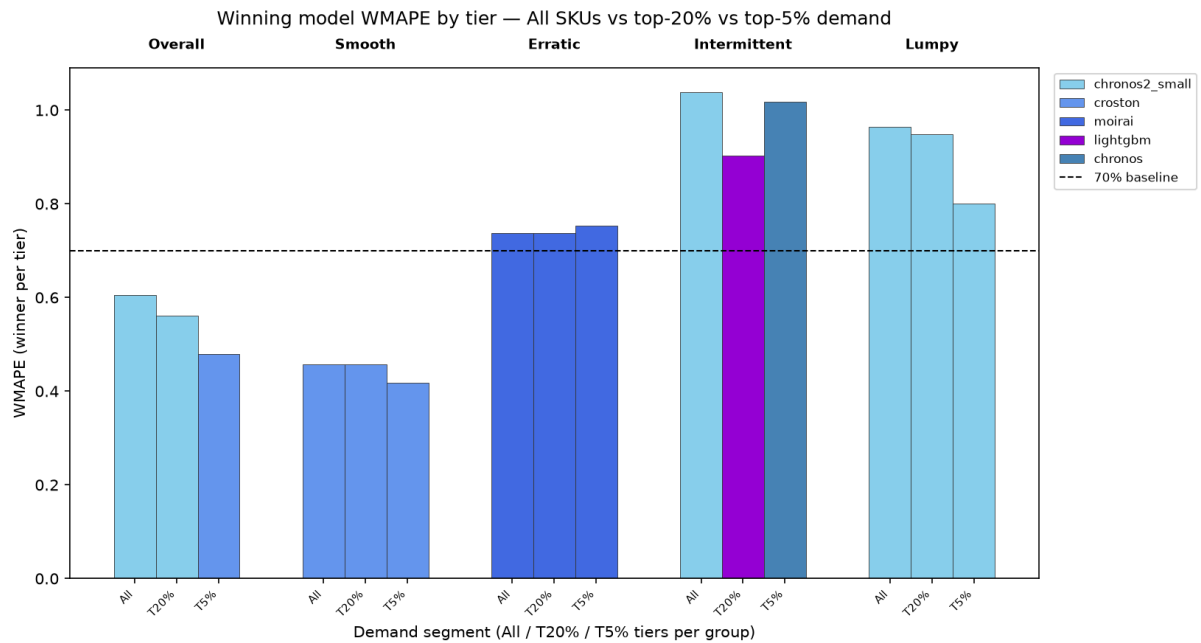


Figure 1. Winning-model WMAPE across all-SKU, top-20% and top-5% demand tiers

Conclusion

This project converted a sparse, commercially important forecasting problem into a reproducible decision-support workflow. The finding that 81.4% of SKUs were intermittent, while smooth items generated most demand, justified segmentation by demand behaviour and commercial volume.

The standardised rolling backtest identifies Chronos2 Small as the overall pipeline winner: it achieved the lowest WMAPE across all eligible SKUs (approximately 61%) and the top 20% (56%). Croston led the top 5% (48%) and narrowly won smooth demand. Because smooth-model performance was tightly clustered, that choice is not operationally decisive. However, intermittent, erratic and lumpy segments remained above the 70% benchmark, so their point forecasts require additional inventory safeguards.

The recommended deployment is therefore a monitored multi-model approach, with Chronos2 as the portfolio benchmark and alternative segment or tier winners used only when their advantage is stable. The 64-month history cannot rule out chance wins or overfitting, and the current SKU-level comparison may overstate Naive because it outputs whole units while other models produce fractions. All forecasts should be rounded to non-negative integers and revalidated before deployment. Where no model beats 70%, Croston should anchor safety-stock calculations, combined with lead times, service-level targets, inventory costs and planner review. This gives Inchcape a transparent basis for improving availability without relying on false precision.

Appendices

Appendix A – Data profile

Monthly collision demand

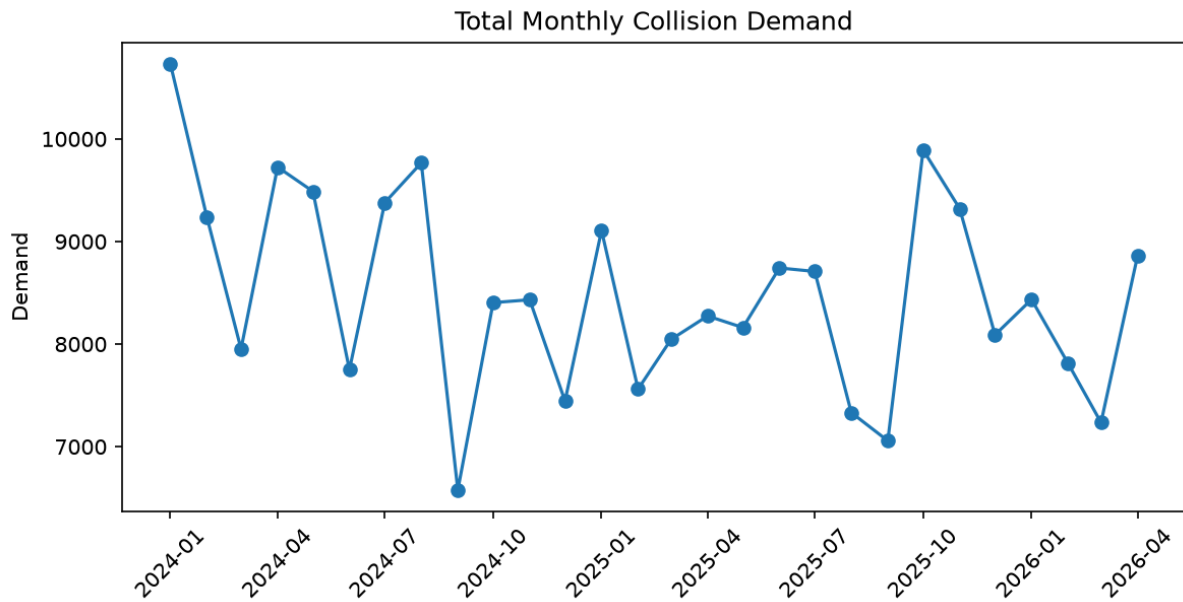


Figure 2. Monthly collision demand over the available history

Active SKUs per month

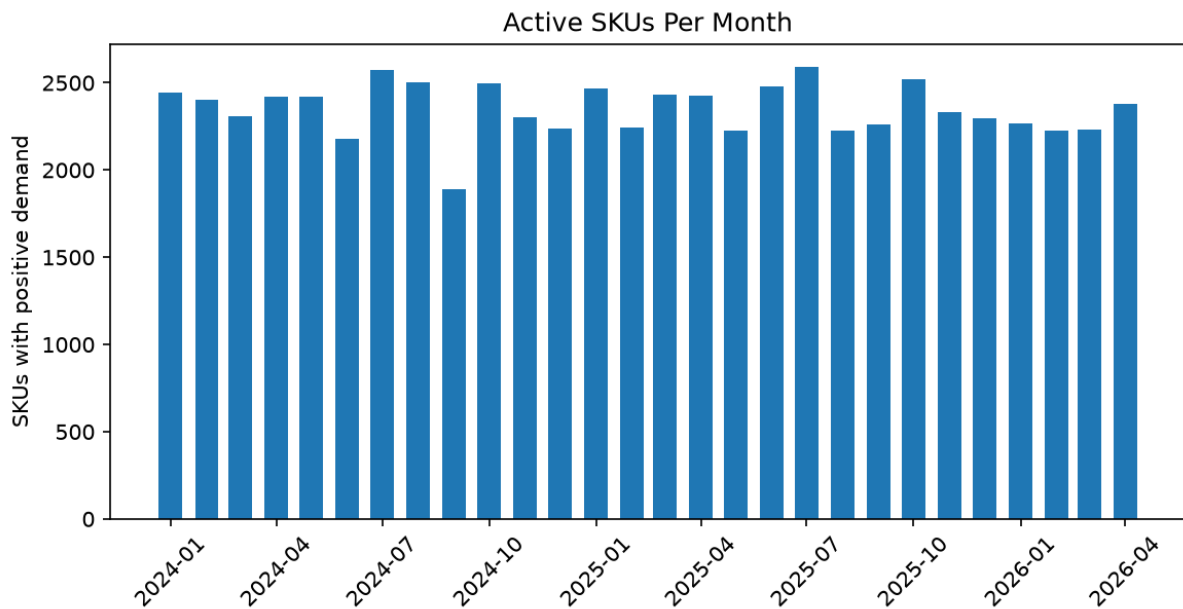


Figure 3. Number of active collision-part SKUs per month

Demand distribution across the top 5% and 20%, by demand type

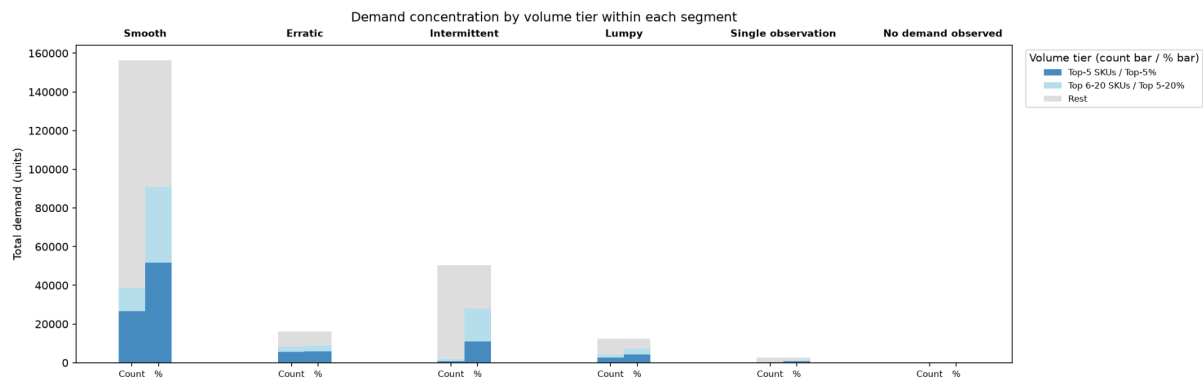


Figure 4. Demand distribution across the top 5% and top 20% of SKUs, by demand type

Backtest eligibility and demand by demand type

Most smooth SKUs had sufficient history for rolling evaluation, whereas fewer than half of intermittent SKUs met the backtest requirement. This limits the certainty of conclusions for short-history sparse items.

Backtest eligibility required at least 12 months of history before the first origin, in addition to the 12-month evaluation horizon and three-month information lag.

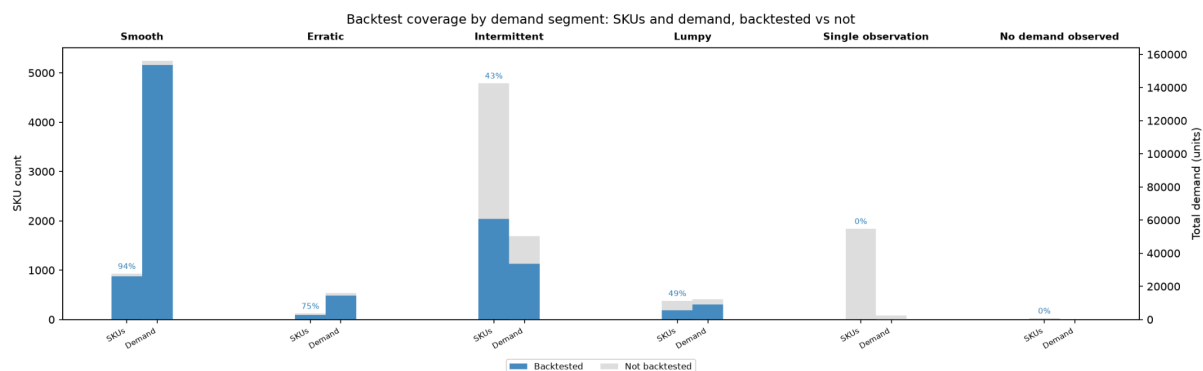


Figure 5. Backtest eligibility and demand volume by demand type

Representative SKUs by demand type

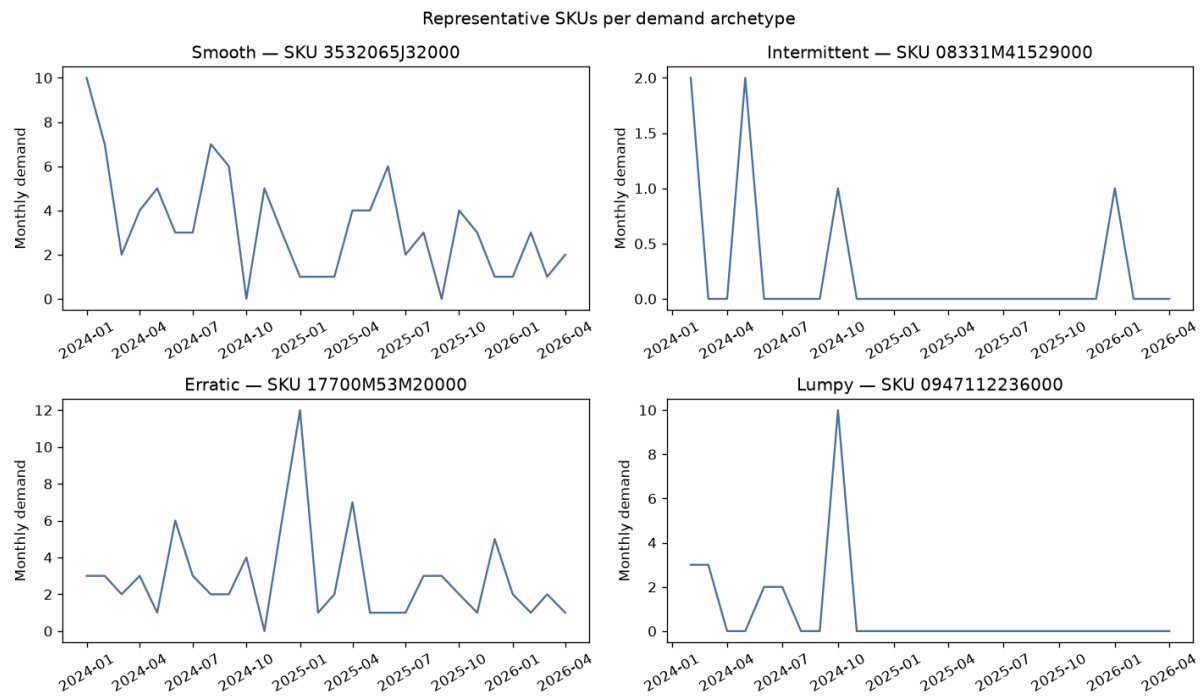


Figure 6. Representative demand histories for smooth, intermittent, erratic and lumpy SKUs

Appendix B – Model winners by demand type and volume tier

All-model performance by demand type

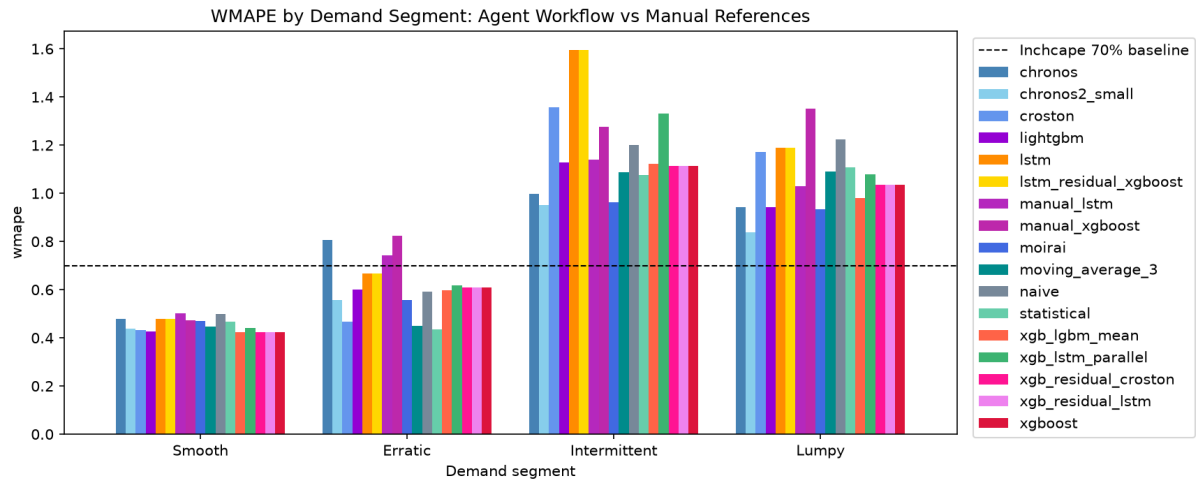


Figure 7. WMAPE performance of all models by demand type

Model winners by demand type: all backtested SKUs

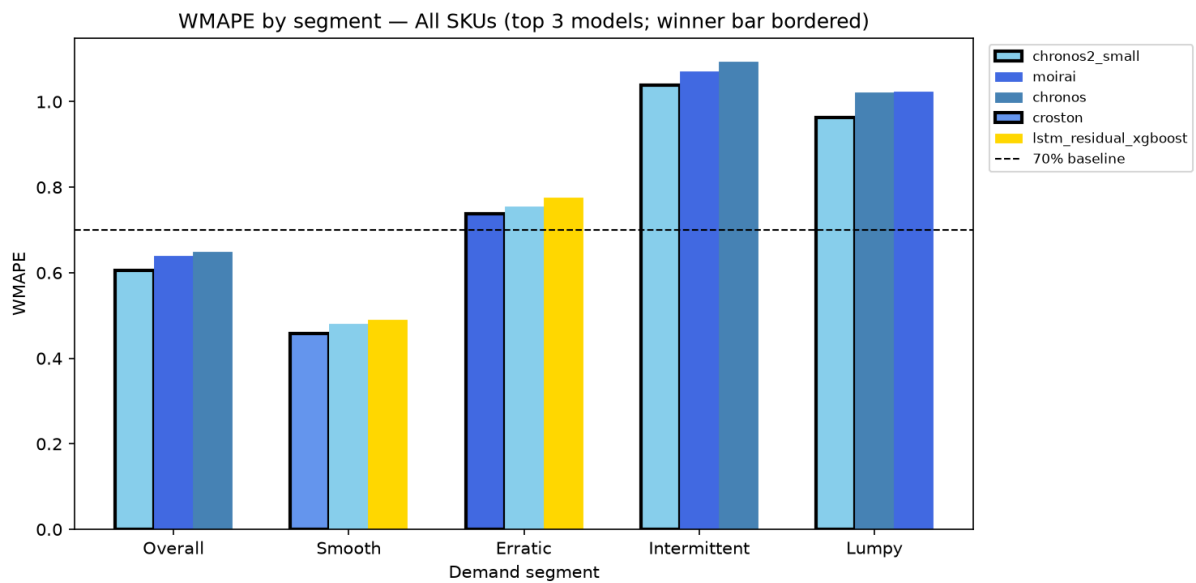


Figure 8. Winning models by demand type across all backtested SKUs

Model winners by demand type: top 20% by demand

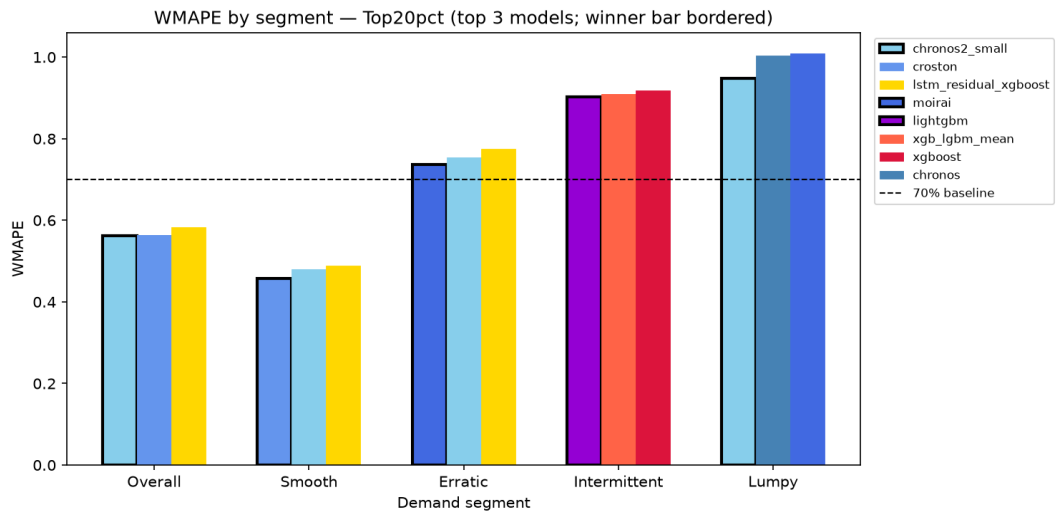


Figure 9. Winning models by demand type for the top 20% of SKUs by demand

Model winners by demand type: top 5% by demand

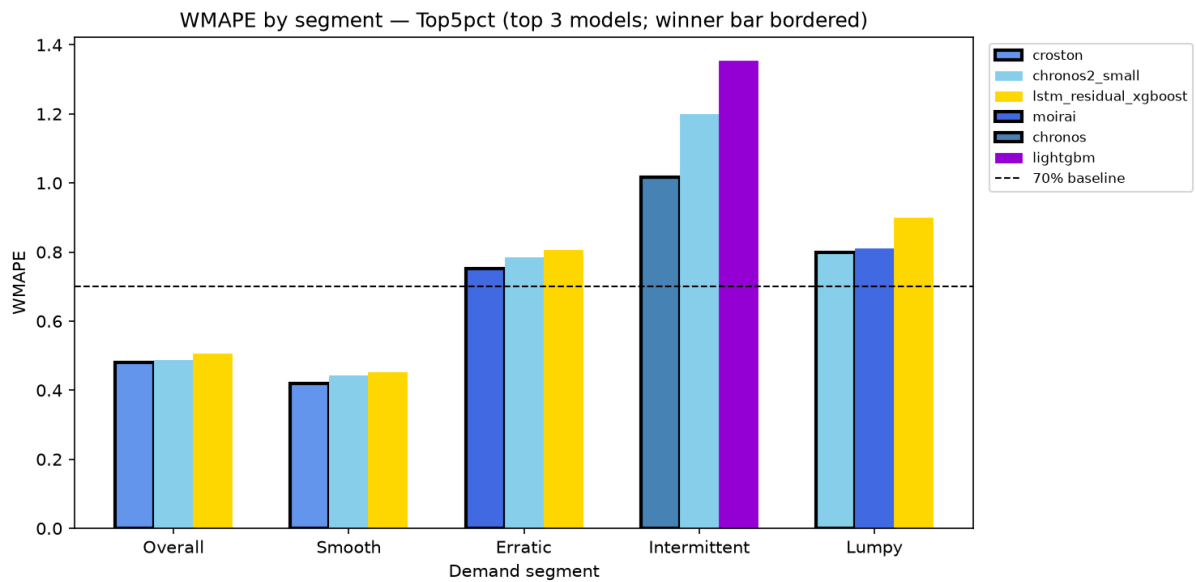


Figure 10. Winning models by demand type for the top 5% of SKUs by demand

Winner comparison across all-SKU, top-20% and top-5% tiers

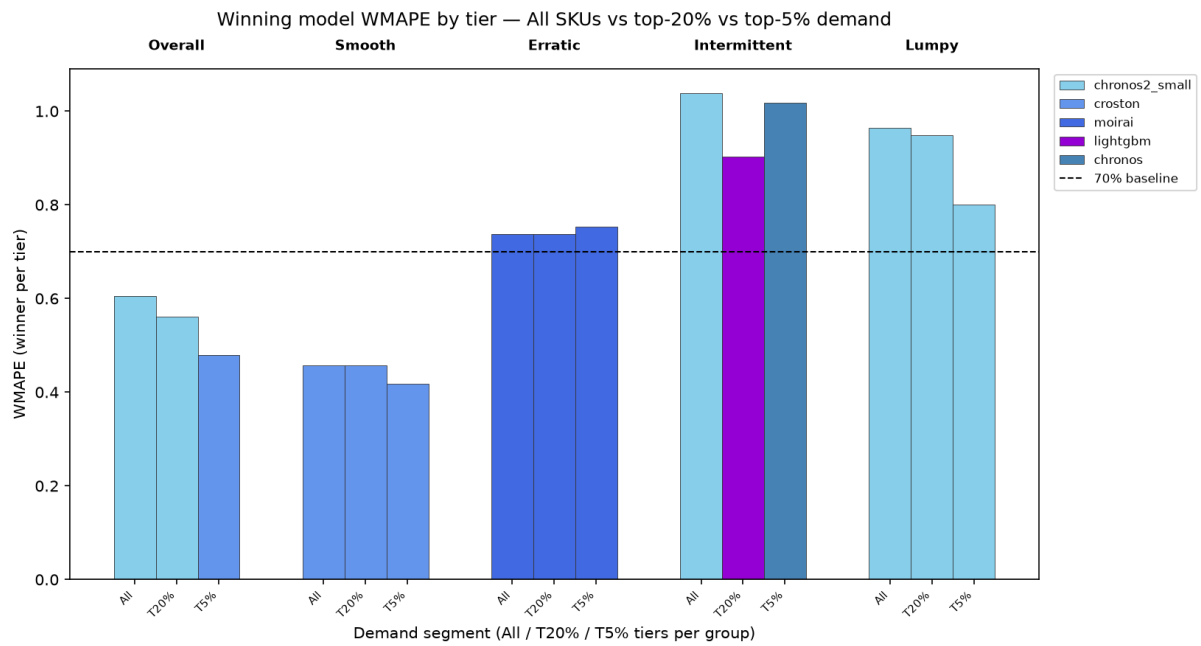


Figure 11. Winning-model WMAPE across all-SKU, top-20% and top-5% tiers

Appendix C – WMAPE metrics

WMAPE definitions: pooled, rolling and employer-requested windows

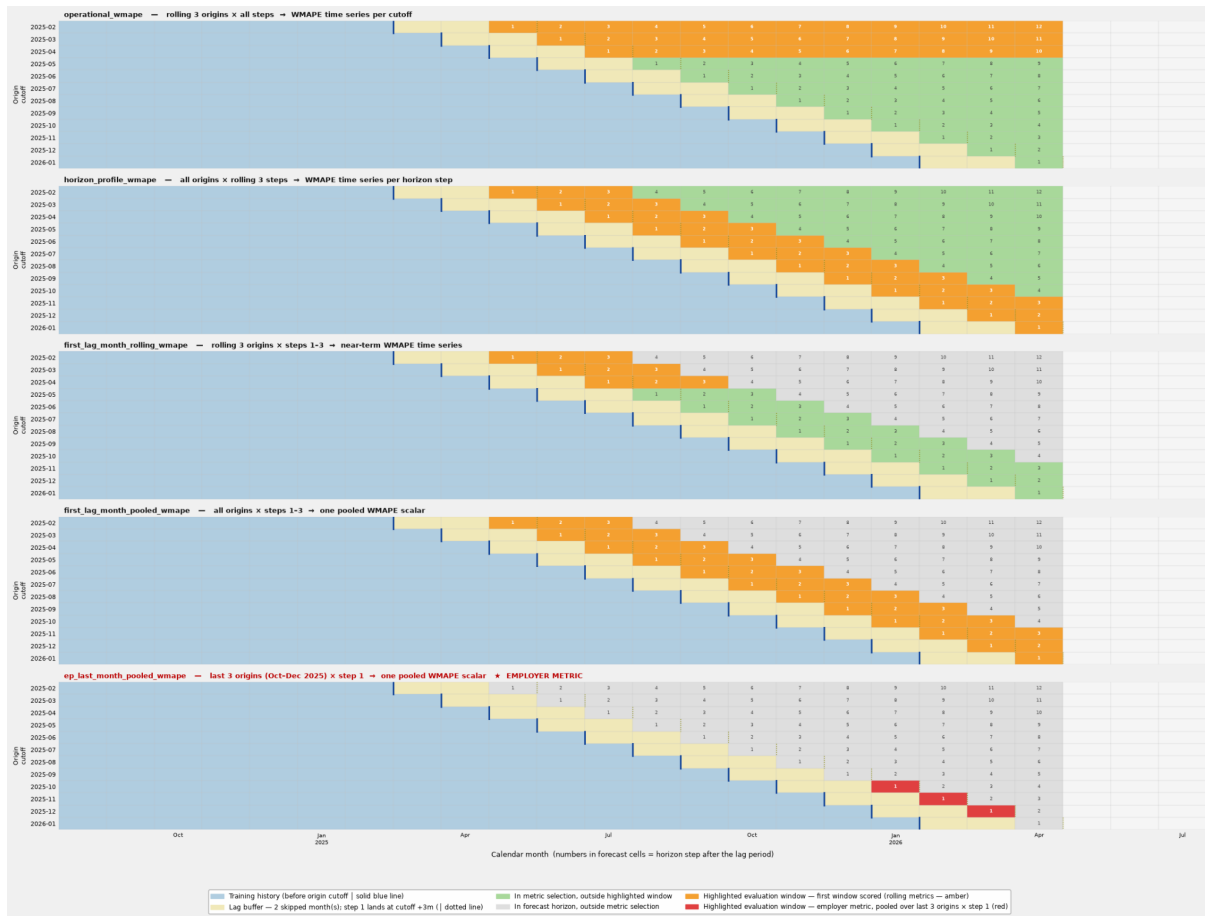


Figure 12. Comparison of pooled, rolling and employer-requested WMAPE windows

Operational WMAPE stability by backtest origin

Most leading models remained broadly stable across backtest origins.

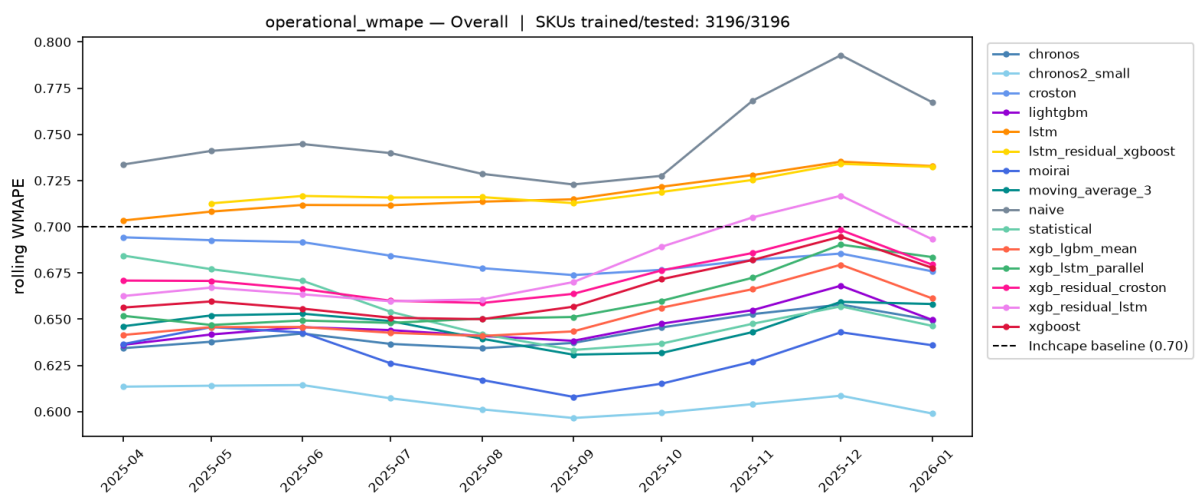


Figure 13. Overall operational WMAPE across rolling backtest origins

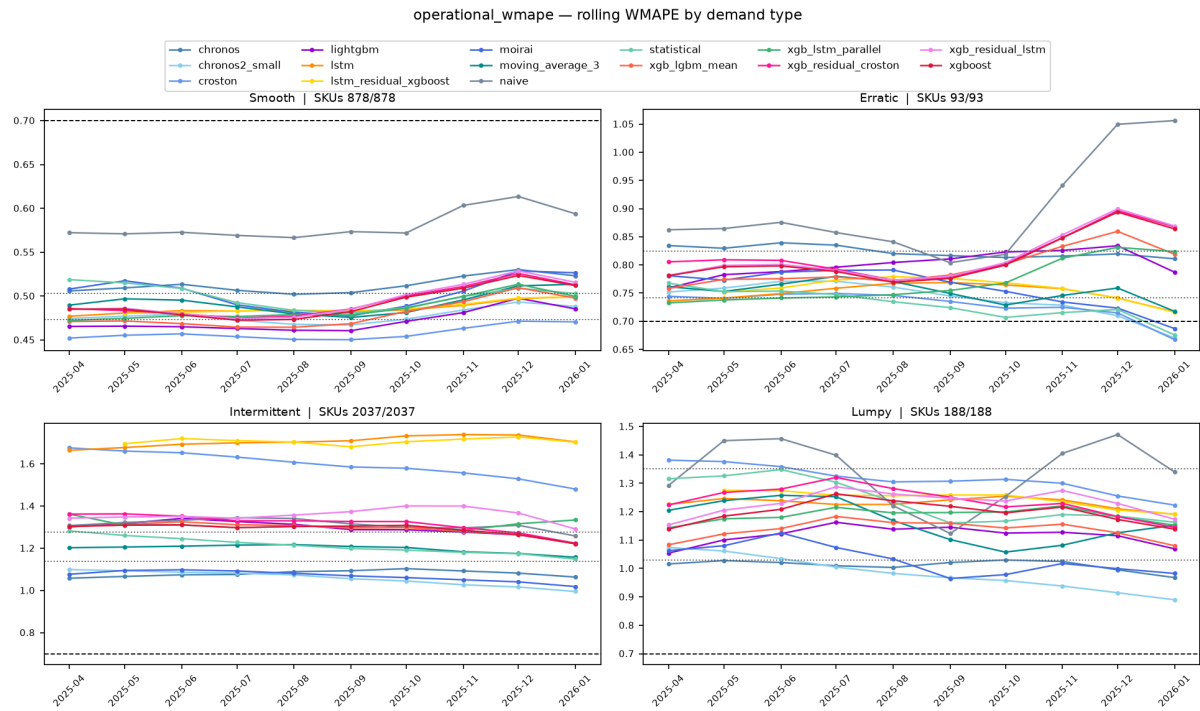


Figure 14. Operational WMAPE by demand type across rolling backtest origins

WMAPE degradation across the forecast horizon

Smooth and erratic segments deteriorated by approximately 10 percentage points over the 12-month horizon; intermittent and lumpy segments were comparatively stable.

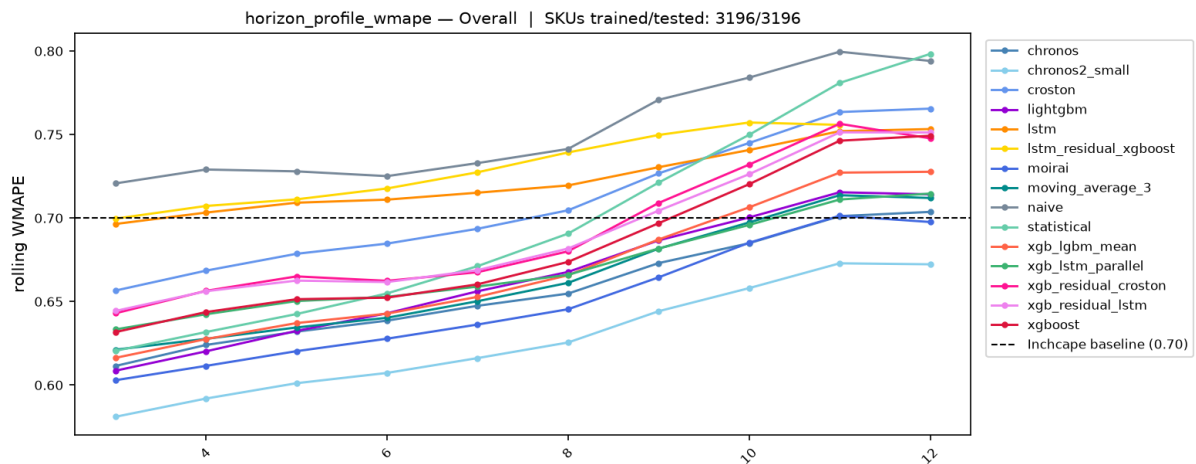


Figure 15. WMAPE degradation by demand type across the forecast horizon

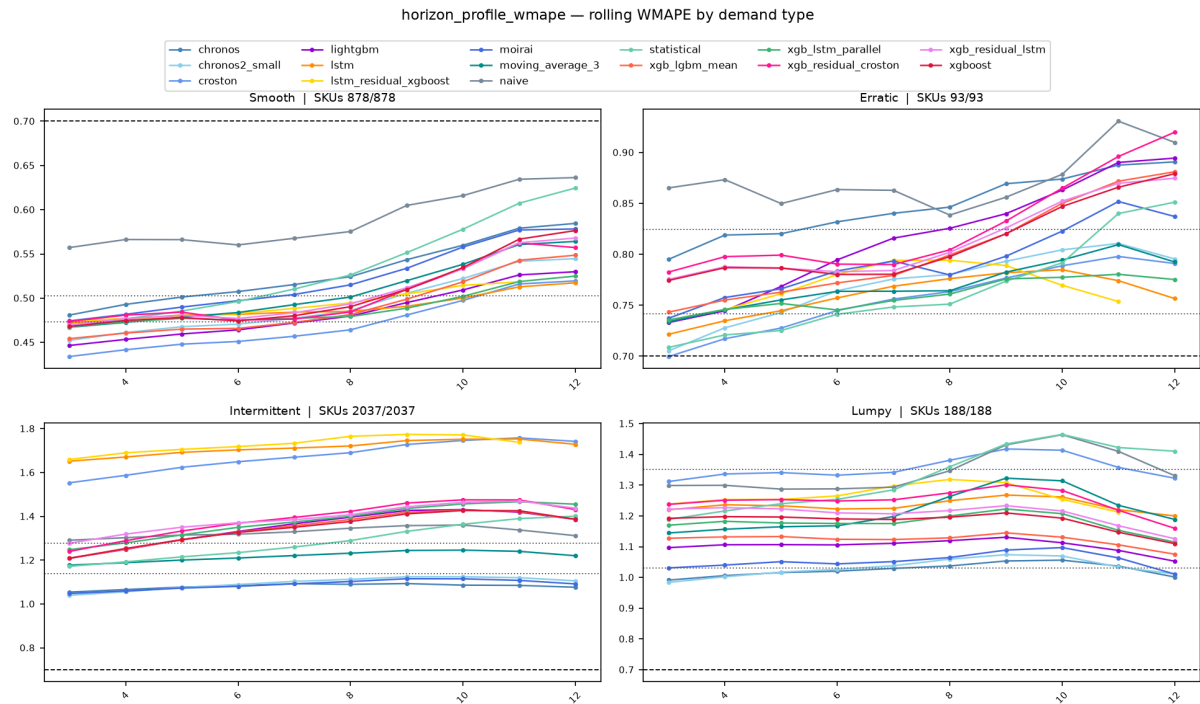


Figure 16. Overall WMAPE profile across the forecast horizon

Overall WMAPE comparison for winning models

The plots compare the distribution of winning-model WMAPE under three aggregation views.

The employer metric, based on the March 2026 prediction window, is generally higher than the first-lag pooled backtest measure but tracks the operational WMAPE pattern.

Operational WMAPE is higher because it includes more distant horizons, where smooth and erratic forecasts lose accuracy.

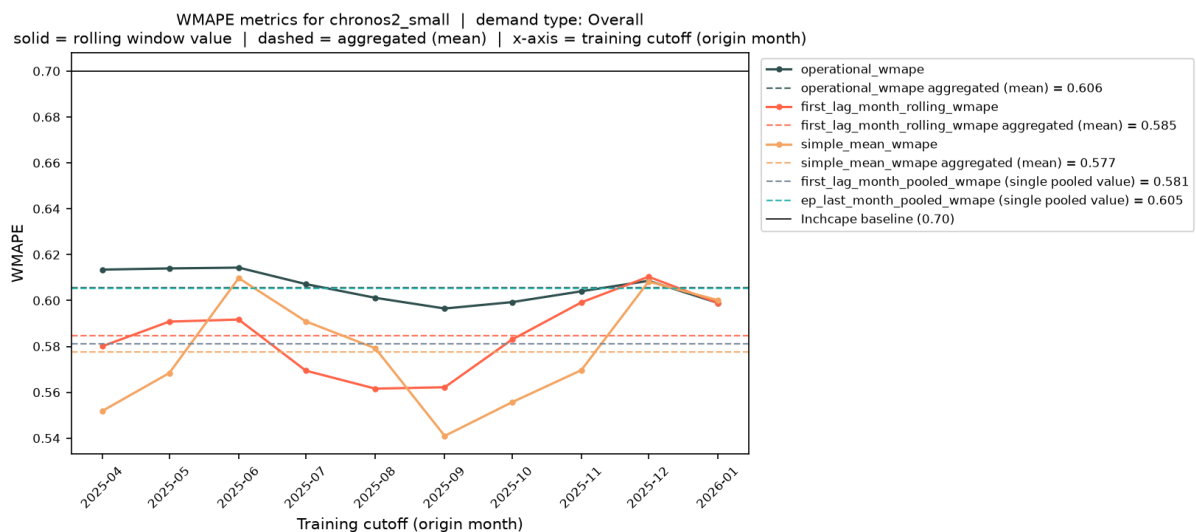


Figure 17. Employer-requested WMAPE comparison for winning models

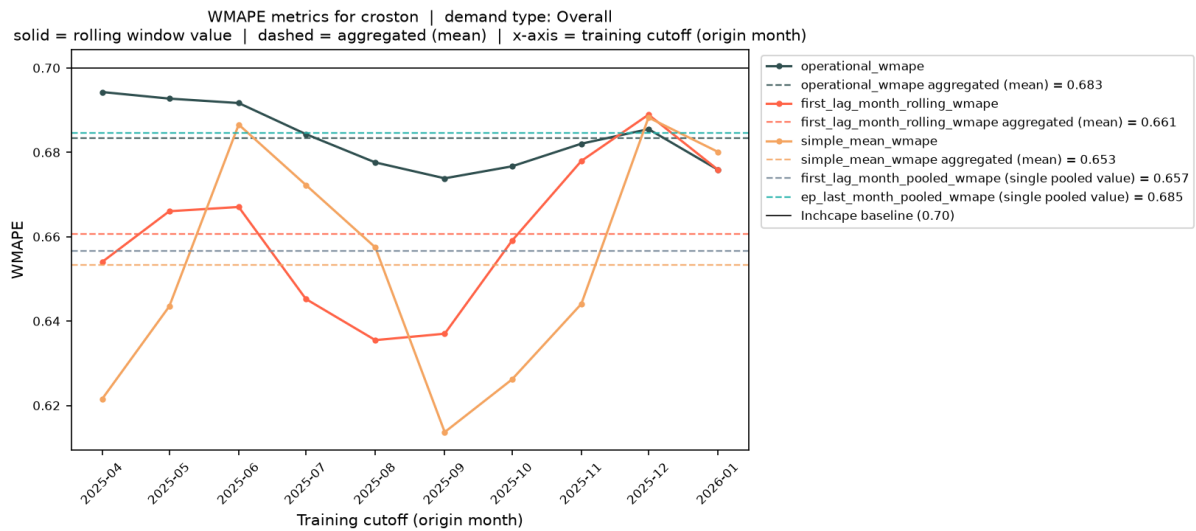


Figure 18. Operational WMAPE comparison for winning models

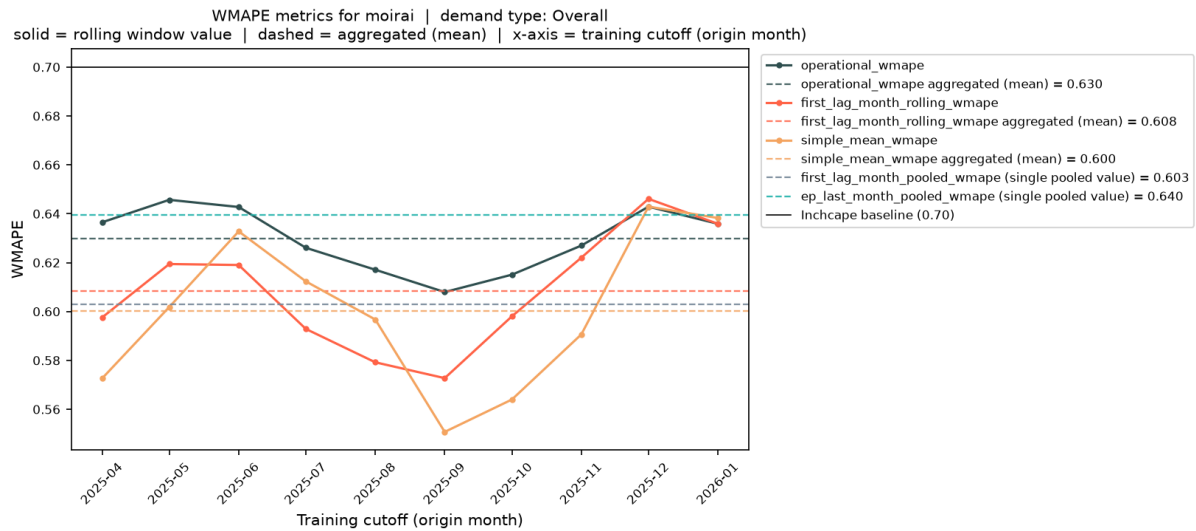


Figure 19. First-lag pooled WMAPE comparison for winning models

Operational WMAPE distribution by model and segment

Boxplots summarise the distribution of operational WMAPE across backtest origins for each model and segment.

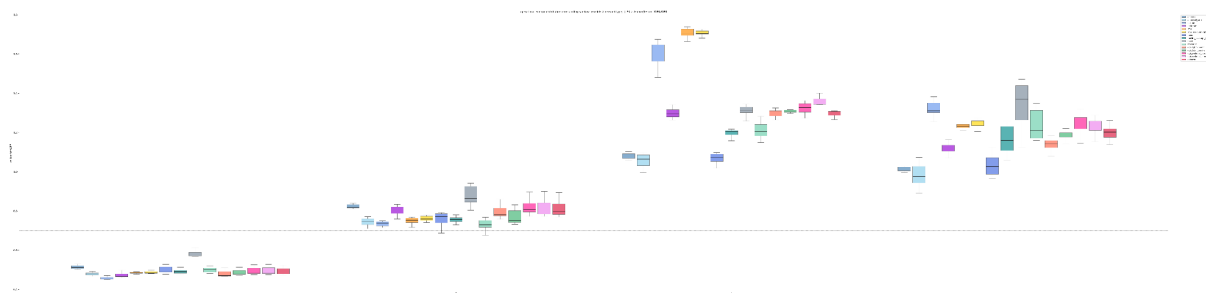


Figure 20. Operational WMAPE distribution by model and demand segment

WMAPE by model for high-volume SKUs

High-demand SKUs generally achieve lower WMAPE across models; the main report therefore focuses on the leading methods within each tier.

Coverage below the 70% WMAPE benchmark is shown for the leading and near-leading models.

Percentage-point improvement by model

Percentage-point improvement is reported relative to the three-month moving-average baseline.

Figure 1 consists of two bar charts. The left chart, titled 'Improvement in WMAPE points', shows the improvement in WMAPE points for various models. The y-axis is labeled 'Δ WMAPE (baseline - model)' and ranges from -0.06 to 0.10. The right chart, titled 'Improvement in WMAPE %', shows the improvement in WMAPE percentage for the same models. The y-axis is labeled 'Δ WMAPE / baseline' and ranges from -100% to 100%.

The models listed on the x-axis for both charts are: chronos2, chronos, moving_average, xgb, lstm, xgb_lstm, xgb_lstm_mean, xgb_lstm_residual, lstm_residual, and xgb_lstm_residual_xgb_lstm.

The following table summarizes the approximate values for each model in both charts:

Model	Improvement in WMAPE points (Δ WMAPE (baseline - model))	Improvement in WMAPE % (Δ WMAPE / baseline)
chronos2	0.095	12%
chronos	0.060	8%
moving_average	0.050	6%
xgb	0.045	5%
lstm	0.025	3%
xgb_lstm	0.020	2%
xgb_lstm_mean	0.015	2%
xgb_lstm_residual	0.005	1%
lstm_residual	-0.010	-1%
xgb_lstm_residual_xgb_lstm	-0.035	-5%

Appendix D – SKU-level analysis

These examples illustrate within-segment variation. Segment winners are selected on aggregate validation performance and will not necessarily be optimal for every individual SKU.

Smooth demand: three best- and worst-performing SKUs

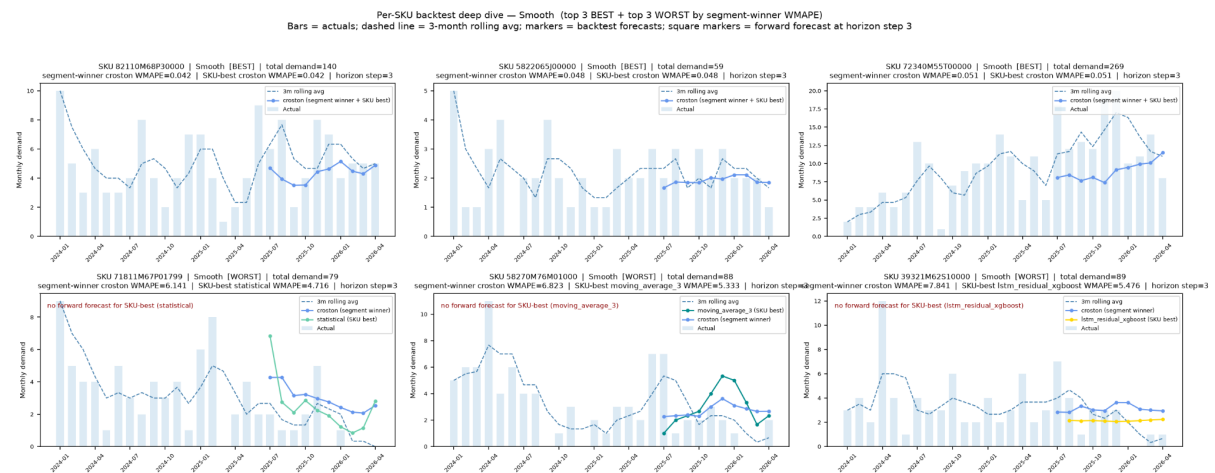


Figure 24. Best- and worst-performing smooth-demand SKUs

Erratic

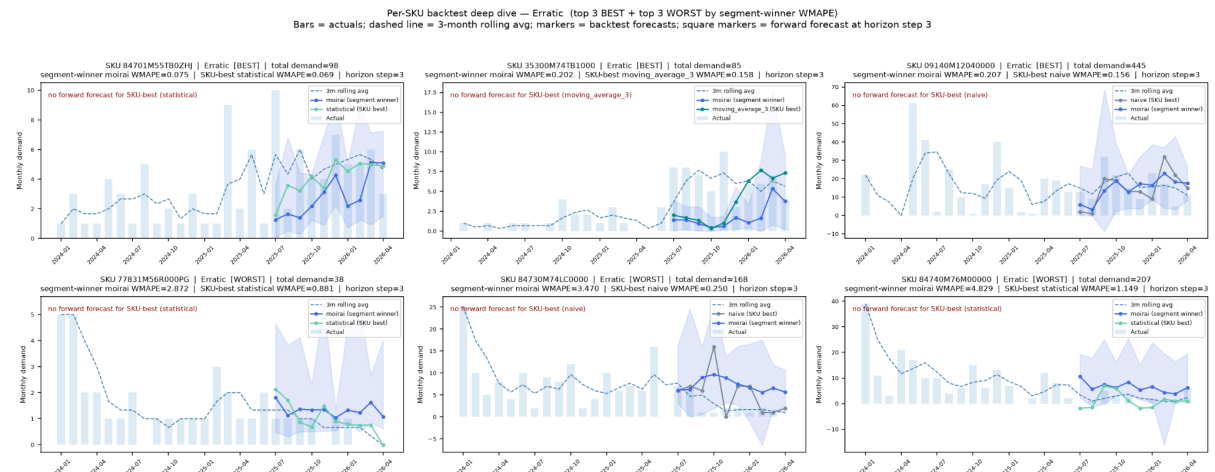


Figure 25. Best- and worst-performing erratic-demand SKUs

Intermittent

Chronos prediction intervals are shown as shaded 10th–90th percentile bands.

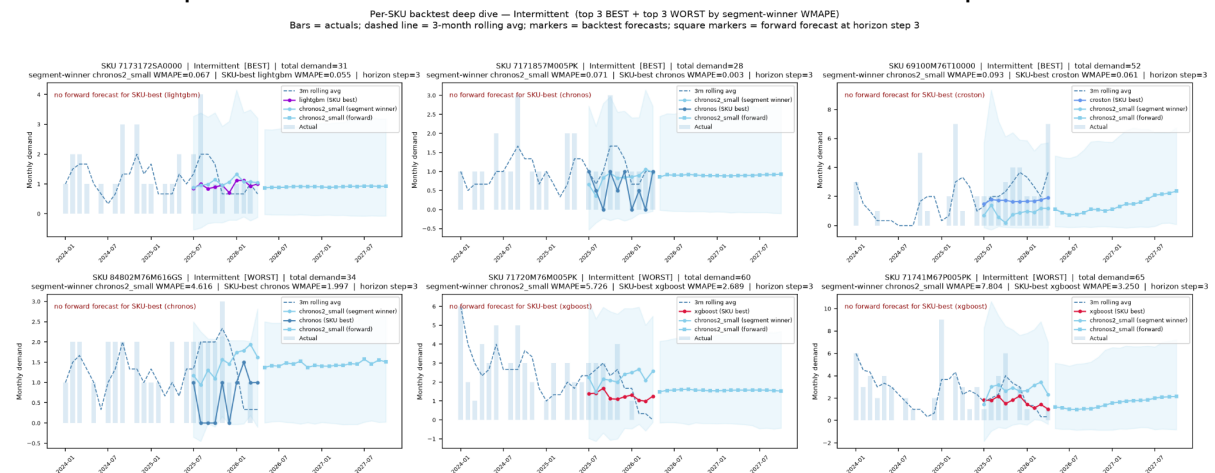


Figure 26. Best- and worst-performing intermittent-demand SKUs

Lumpy

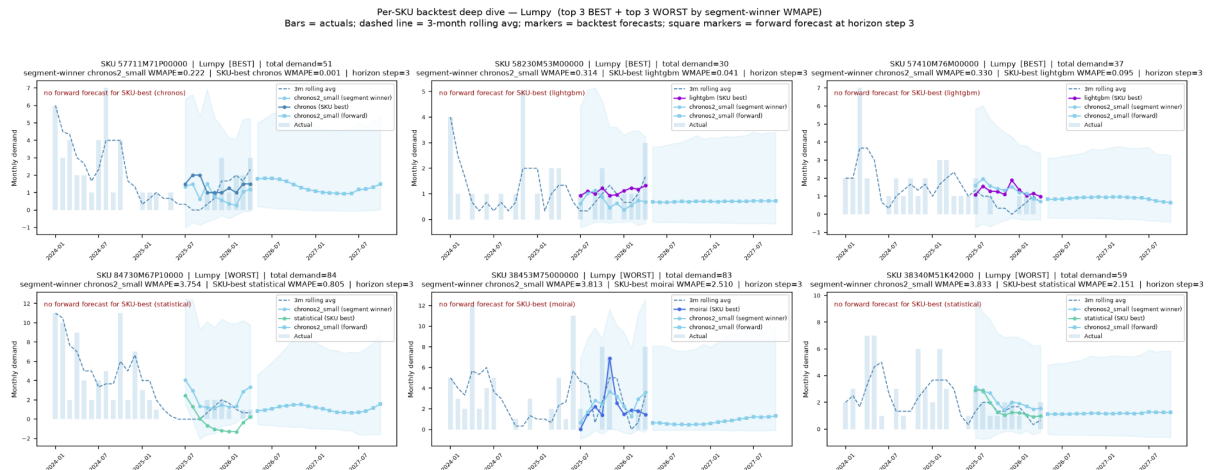


Figure 27. Best- and worst-performing lumpy-demand SKUs

High Volume

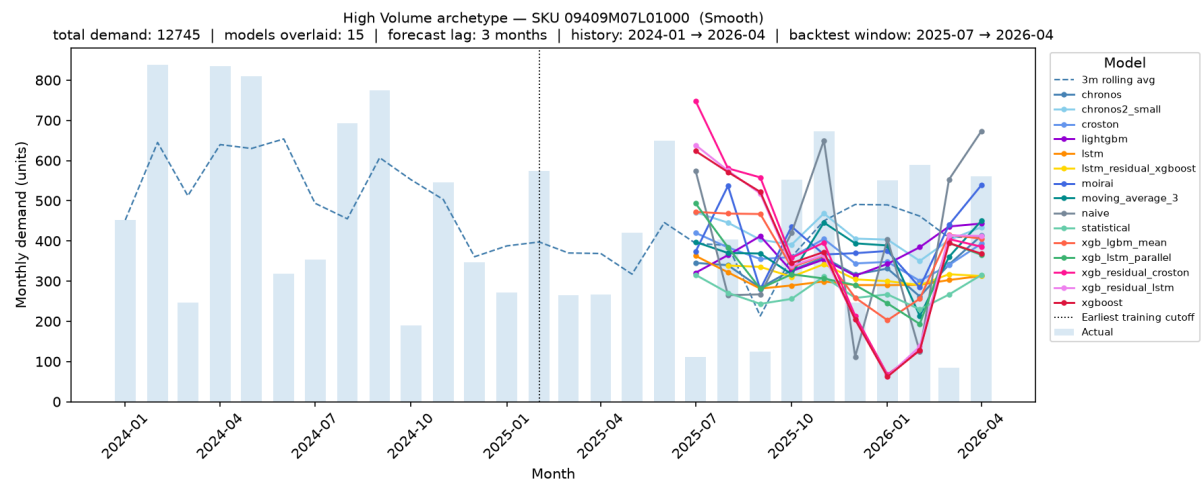


Figure 28. Forecast comparison for selected high-volume SKUs

Agreement between SKU-level and segment-level winners

This diagnostic test whether the segment-level winner also matches the best SKU-level model, providing evidence on whether demand-type segmentation is sufficiently granular.

Appendix E – Forecast outputs

Aggregate 18-month forecast across all SKUs

Consistency control: aggregate forecasts should be generated from the same final pipeline run used for model selection.

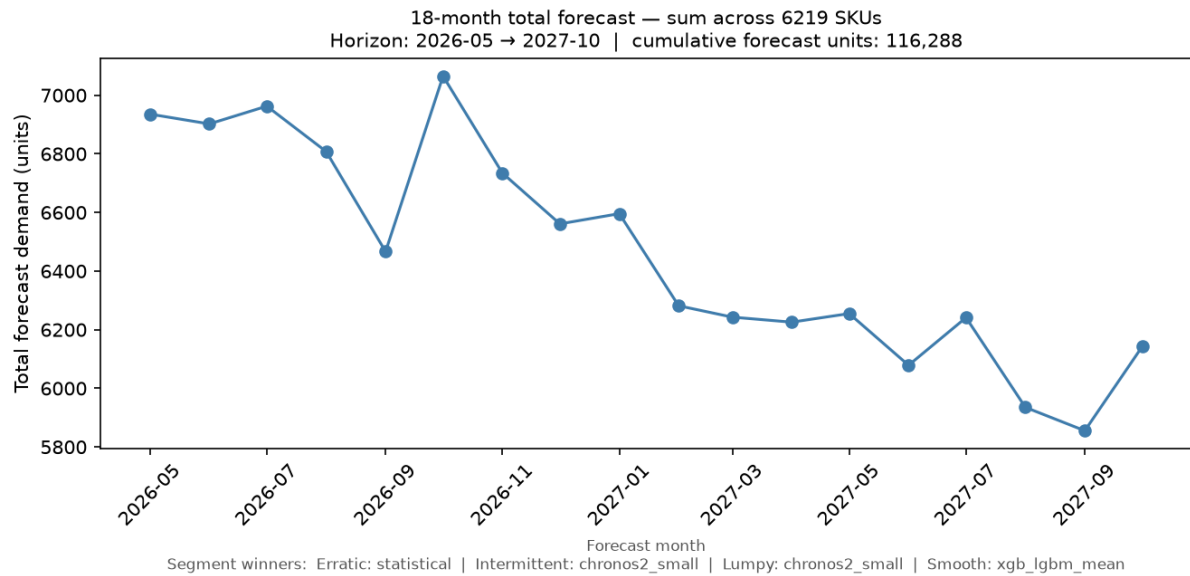


Figure 29. Aggregate 18-month forecast across all collision-part SKUs

Aggregate 18-month forecast by demand type

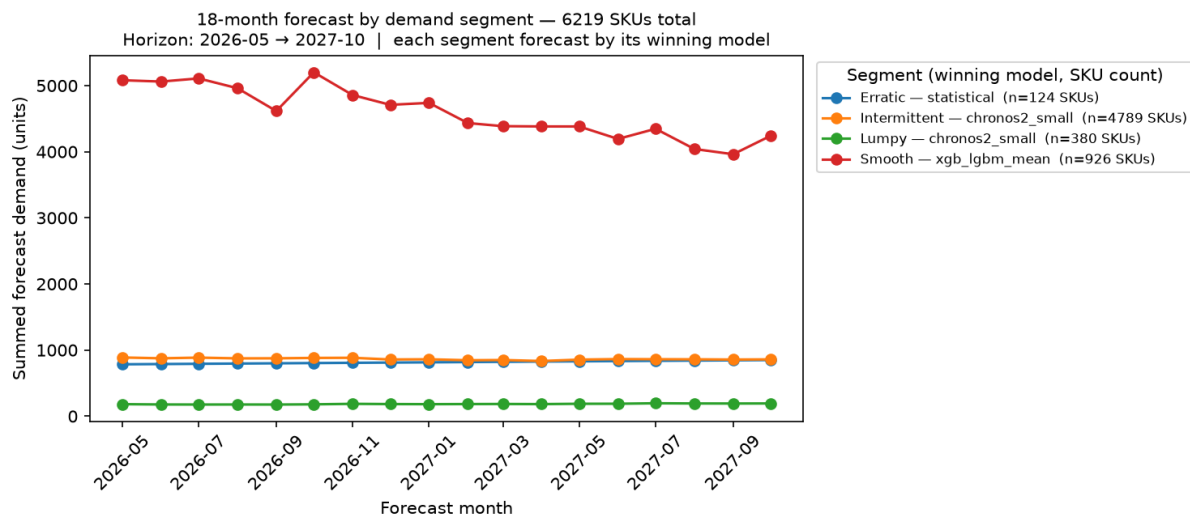


Figure 30. Aggregate 18-month forecast by demand type

Sample SKU forecasts: three per segment

Sample forecasts compare predicted demand with history; production views should also display the relevant backtest trace.

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18-month SKU forecasts by demand type — 12 SKUs across 4 segments
Forecast window: 2026-05 → 2027-10 | dotted vertical line = last actual month; forecast lines extend to the right



Figure 31. Sample 18-month forecasts for selected SKUs from each segment

Appendix F – Exogenous variables

Pearson correlations for global exogenous variables

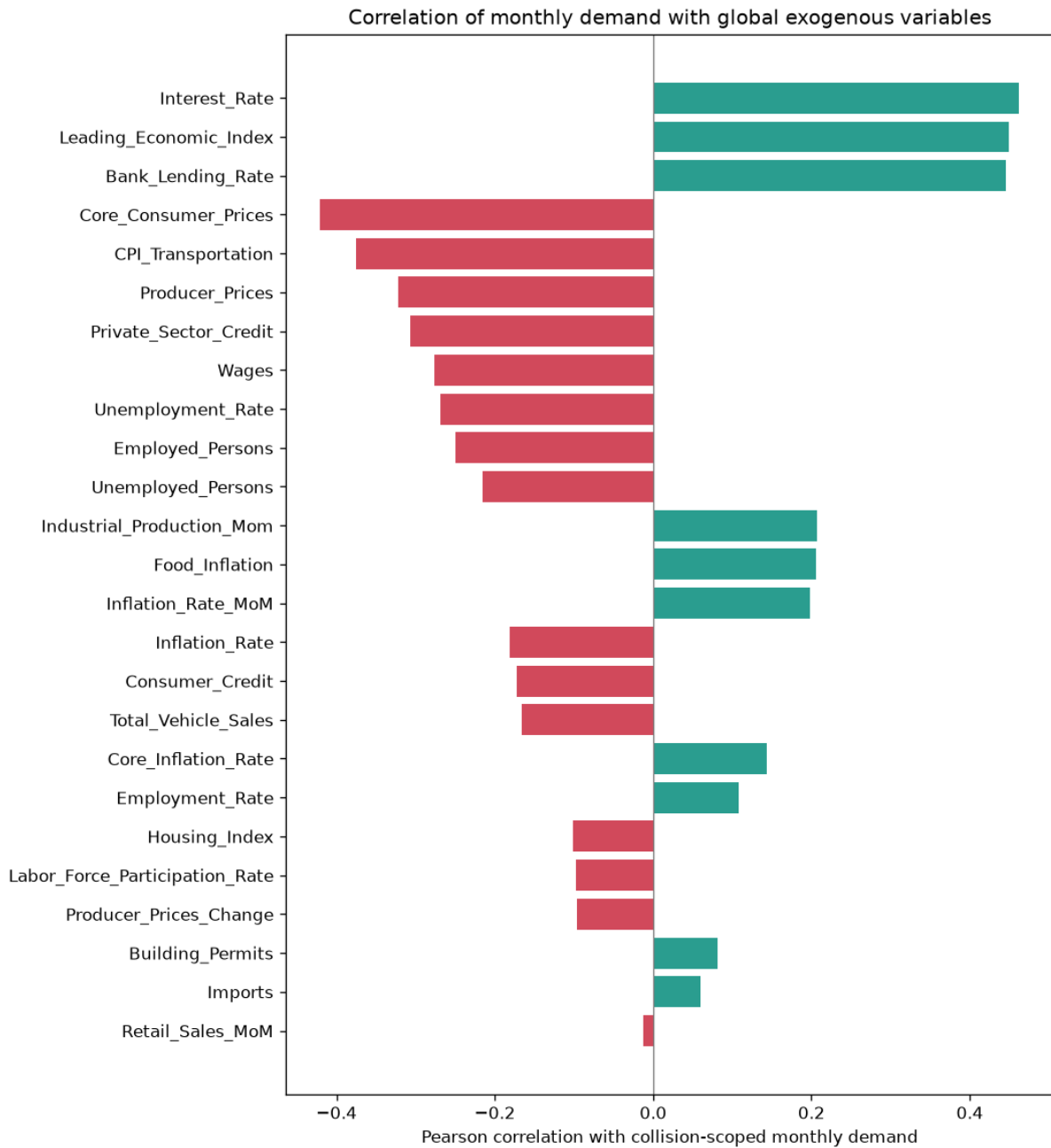


Figure 32. Pearson correlations between exogenous variables and collision-part demand

Appendix G – Models and backtesting

Models tested

Model	Description
naive	Last observed actual (3 months earlier)
moving_average_3	Moving average (window size = 3)
xgboost	Feature-based gradient boosting (XGBoost)
lightgbm	Feature-based gradient boosting (LightGBM)
croston	Croston / SBA method for intermittent demand
statistical	Classical time-series model (Theta / SES / SARIMA)
chronos	Zero-shot foundation model forecast (Amazon Chronos)
chronos2_small	Zero-shot Chronos-2 foundation model
moirai	Zero-shot foundation model forecast (Salesforce Moirai)
lstm	Per-demand-type LSTM neural network (non-negative point forecaster)
residual_lstm	Signed-residual LSTM (linear head + MSE loss) for serial ensembling
xgb_lgbm_mean	Parallel ensemble (mean) of: xgboost, lightgbm
xgb_residual_croston	Serial residual ensemble: xgboost + croston(residuals)
xgb_lstm_parallel	Parallel ensemble (weighted_mean_search) of: xgboost, lstm
xgb_residual_lstm	Serial residual ensemble: xgboost + residual_lstm(residuals)
lstm_residual_xgboost	Serial residual ensemble: lstm + xgboost (residuals)

Table 2. Forecasting model registry

Feature engineering

Feature family	Variables
Demand lags	lag_1, lag_2, lag_3, lag_6, lag_9, lag_12, lag_15
Known-history lags	lag_1_known, lag_2_known, lag_3_known, lag_6_known, lag_9_known, lag_12_known, lag_15_known
Rolling statistics	rolling_mean_3/6; rolling_std_3/6; rolling_max_3/6/12
Sparsity and recency	zero_share_6/12; months_since_last_demand; demand_history_known
Calendar signals	month_of_year; quarter; year

Table 3. Engineered feature matrix

Appendix H – Pipeline

The pipeline diagram is also available in Markdown/Mermaid format in the project repository:

https://github.com/alejandrol06/D8_solutions/blob/Scotts-branch/scripts/pipeline_diagram/pipeline_diagram.md

Autonomous Forecasting Orchestrator — Agent Sequence

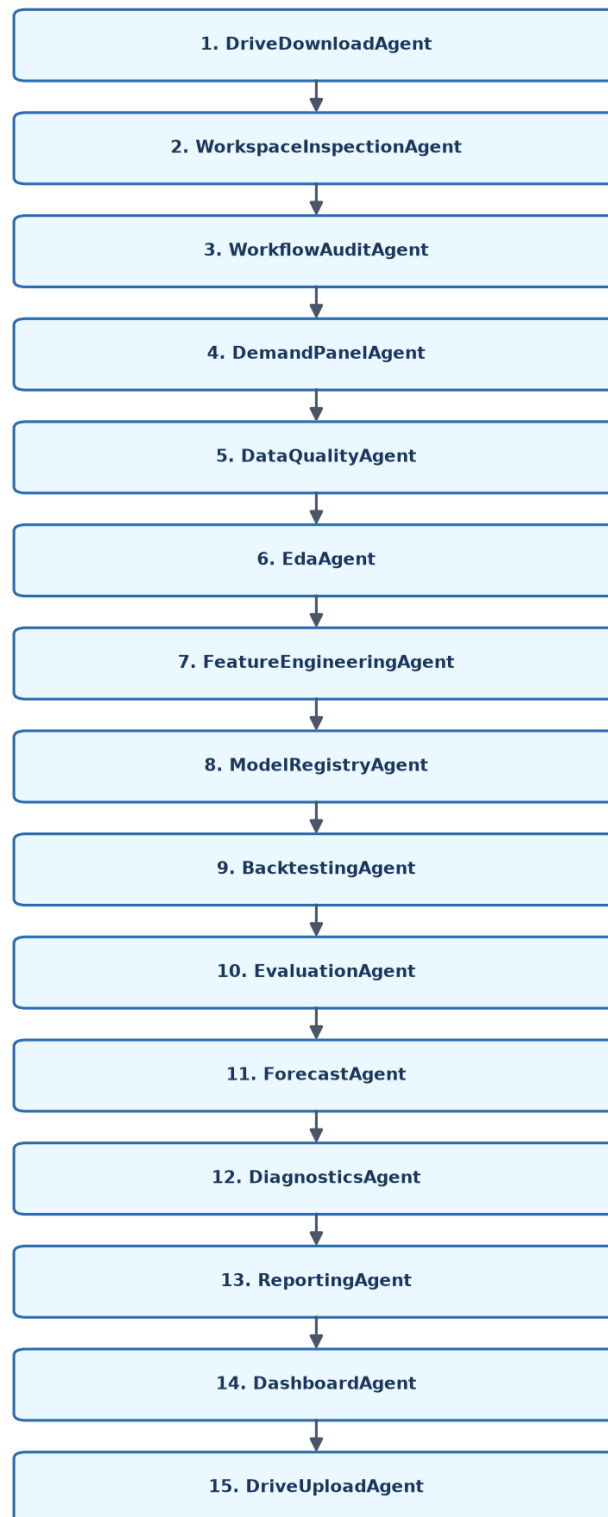


Figure 33. Autonomous forecasting orchestrator and agent sequence